

A Stopping Target Optimized at the *Deployed* Foil Pitch

9D shape search, foil spacing fixed to the baseline design

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Mu2e — autoresearch / closed-loop Bayesian optimization

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Result in one slide

Hold the foil-to-foil **gap at the deployed 22.2 mm** — so the support and frame engineering carry over — and let Bayesian optimization choose everything else. The result beats the deployed target by +32% $S/\sqrt{B}$ and beats the 6D champion on **both** objectives at once: +4.9% significance at −19% **detector damage**.

- **83 evaluations** (13 transplanted seeds + 70 new), **zero failures**; a fourth campaign is running, so the front is still moving.
- **Record** foilspfbp03R03_00: **4.30** at 8.37×10^{-7} — vs. the deployed target's 3.26 at 6.85×10^{-7} , i.e. +32% $S/\sqrt{B}$ for +22% **damage**.
- It **strictly dominates** the previous record (bp01R00_03, 4.29 at 1.23×10^{-6}): same significance, **32% less damage**, a completely different shape.
- **And damage need not be paid at all**: bp04R00_02 reaches +19% $S/\sqrt{B}$ at −13% **damage** — strictly better than the deployed target on *both* axes. **24 of 83** designs do this.
- Damage frontier spans **22×**, floor 5.6×10^{-8} MeV/POT, **28 Pareto points**.

All numbers Run1Bap-scale on the corrected stack (IPA-position fix, strict zero-overlap preflight).

What we optimize — and what we deliberately do not

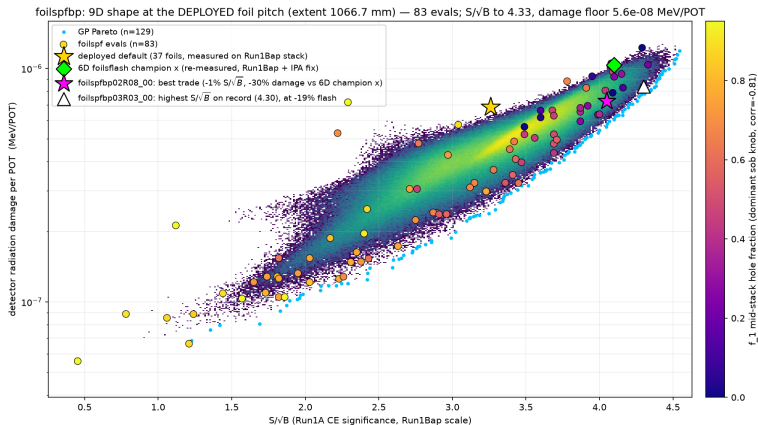
9-D shape at fixed pitch: 49 foils generated from $K=3$ quadratic control points for outer radius, half-thickness and hole fraction. **Extent is pinned** at $48 \times 22.222 = \mathbf{1066.7}$ mm — the deployed foil spacing.

knob	range	
$r_{\text{Out}_{\{0,1,2\}}}$	50–120	mm
$h_{\text{T}_{\{0,1,2\}}}$	0.01–0.15	mm
$f_{\{0,1,2\}}$	0–0.95	hole fraction
extent	1066.7	fixed

objective	direction
$S/\sqrt{B}$ (Run1A CE)	maximize
detector radiation damage per POT	minimize

Why fix the pitch. A designed extent scan (10 evaluations, two shapes, 800–2000 mm) showed $S/\sqrt{B}$ flattening by 1400–1700 mm: the last 300 mm buys +0.9% significance for +8% damage. Length is a settled question — so spend the search on shape, inside an envelope the hardware already supports.

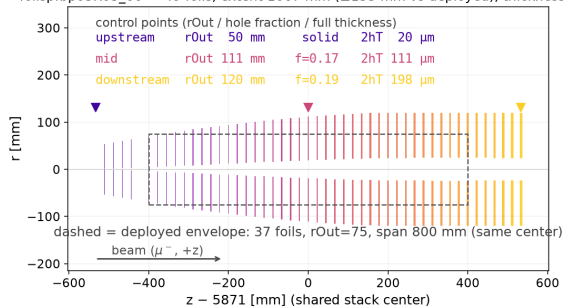
The landscape



83 evaluations, GP cloud over the 9D shape box. Gold star = deployed default; green diamond = the 6D foilsflash champion geometry re-measured on this stack; **white triangle** = the record, sitting *to the right of and below* the green diamond — better on both axes; magenta star = the lowest-damage point still beating the deployed target. Colour = mid-stack hole fraction, the dominant knob (corr **-0.81**).

Anatomy of the record

foilsfpfbp03R03_00 — 49 foils, extent 1067 mm (± 133 mm vs deployed), thickness $\times 30$



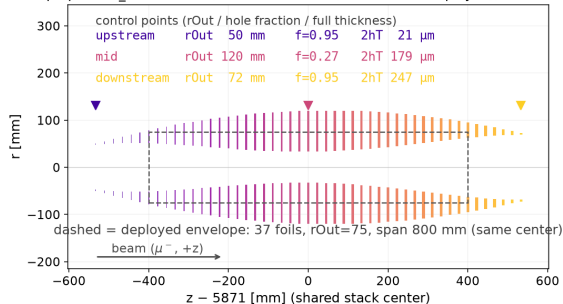
design	$S/\sqrt{B}$	damage
bp03R03_00 (record)	4.30	0.84
bp01R00_03 (old rec.)	4.29	1.23
bp02R03_00	4.09	0.79
bp02R08_00	4.05	0.73
6D champion $\times$ (re-meas.)	4.10	1.04
deployed default	3.26	0.69

damage in 10^{-6} MeV/POT, all on the Run1Bap stack

- **Grows outward, stays nearly solid.** rOut 50/111/120 mm with holes almost shut ($f = 0/0.17/0.19$) — the reverse of the old record, which shrank downstream to rOut 50 and opened an upstream hole.
- Thickness ramps $20 \rightarrow 111 \rightarrow 198 \mu\text{m}$: thin where the beam is still intact, thick where it is spent.
- Growth is **symmetric** about the shared centre $z_0 = 5871$: ± 133 mm beyond the deployed span, dashed = deployed envelope.

Anatomy of the damage-free point

foilsfpfbp04R00_02 — 49 foils, extent 1067 mm (± 133 mm vs deployed), thickness $\times 30$



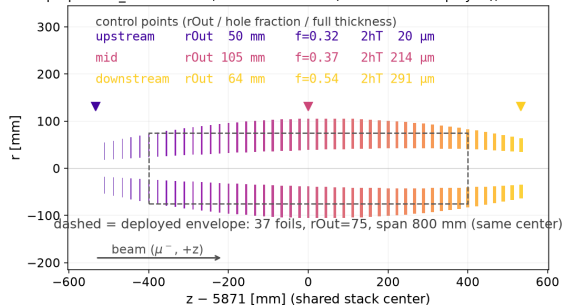
design	$S/\sqrt{B}$	damage
bp04R00_02	3.87	0.59
deployed default	3.26	0.69
<i>difference</i>	+19%	-13%

damage in 10^{-6} MeV/POT, Run1Bap stack

- **Better on both objectives** than the target we deploy today — no trade at all. **24 of 83** evaluated designs do this.
- **A wider bore, and one that varies.** The deployed target is already annular, but at a *constant* 21.5 mm hole; this one runs **33–69 mm** — 1.5–3 $\times$ wider — and profiles it along z .
- Carries **216 cm³** of aluminium — 3.4 $\times$ the deployed 63 cm³ — but only **1.5%** of it inside $r < 40$ mm, against the deployed target's 22%.
- More stopping mass *and* less flash: the mass moved off the beam axis.

Anatomy of the best point at deployed-level damage

foilsfpbp02R08_00 — 49 foils, extent 1067 mm (± 133 mm vs deployed), thickness $\times 30$



design	$S/\sqrt{B}$	damage
bp02R08_00	4.05	0.73
deployed default	3.26	0.69
<i>difference</i>	+24%	+6%

damage in 10^{-6} MeV/POT, Run1Bap stack

- **The most significance available at deployed-level damage:** highest $S/\sqrt{B}$ of any design within 10% of the default's flash.
- Buys +5 points of $S/\sqrt{B}$ over the damage-free design for 19 points of damage — still only +6% against today's target.
- **Tighter bore:** 16–42 mm, straddling the deployed 21.5 mm, vs the damage-free point's 33–69 mm. Core fraction **3.1%** vs 1.5%.
- 212 cm³ of Al, $3.4\times$ the deployed 63 cm³, thickest downstream (291 μm) where the beam is spent.

What actually drives the damage — everything vs. the deployed target

design	$S/\sqrt{B}$	vs default	damage	vs default	Al [cm ³]	Al in $r < 40$ mm
deployed default (37 foils)	3.26	—	0.69	—	63	22.0%
bp04R00_02 (dominates)	3.87	+19%	0.59	−13%	216	1.5%
bp02R08_00	4.05	+24%	0.73	+6%	213	3.1%
bp02R03_00	4.09	+25%	0.79	+15%	166	9.2%
bp03R03_00 (record)	4.30	+32%	0.84	+22%	206	9.6%
bp01R00_03 (old record)	4.29	+32%	1.23	+79%	153	22.0%

damage in 10^{-6} MeV/POT; all measured on the Run1Bap stack; rows ordered by damage

Every optimized design carries **2.4–3.9×** the default's aluminium — that is how it buys the significance. What varies by **13×** is the damage *penalty*: −13% to +79%. It tracks the core column, not the mass column.

Over all **83** evaluated geometries:

Spearman vs. damage	ρ
total Al volume	+0.57
Al inside $r < 40$ mm	+0.84

- bp01R00_03 tripled the mass and *kept* the default's 22% core fraction → +79% damage. bp04R00_02 carried *more* mass at a 1.5% core → −13%.
- So the lever is **placement, not mass** — and it is strong enough that **23 of 81** evaluated designs beat the deployed target on *both* objectives at once.
- Consistent with Edmonds (DocDB-10898) — flash parents are central, $RMS \approx 24$ mm — found here by search, not by design.

Why the optimum moves with the pitch

The same shape is *not* optimal at every length — transplanting the 2000 mm champion into this envelope leaves performance on the table.

shape at 1066–1100 mm	$S/\sqrt{B}$	hole profile f
transplanted 2000 mm champion (SCANA1100)	4.16	0.43 / 0.12 / 0.86
optimized at this pitch (bp01R00_03)	4.29	0.43 / 0 / 0
record (bp03R03_00)	4.30	0 / 0.17 / 0.19

- +3.4% ($\sim 5.6\sigma$ at $\sigma_{\text{sob}} = 0.6\%$) purely from re-optimizing the shape at fixed length. **Two independent optima found here both shut the downstream hole** the 2000 mm champion opens wide.
- **Mechanism:** at tight spacing a conversion electron re-crosses more neighbouring foils on its way out, so exit-path self-absorption dominates and *open holes stop paying*. At 42 mm pitch the same optimizer opens them wide.
- Consistent with the extent scan, where the length gain itself came from the CE low-momentum tail retreating — a narrower signal window against a cosmic-dominated background.

How fast it converges — and when to stop

campaign	evals	best $S/\sqrt{B}$	outcome
foilspfbp01	20	3.56 $\rightarrow$ 4.29	winner arrived in wave 0
foilspfbp02	20	4.29 $\rightarrow$ 4.29	0 of 20 beat it; 7 front points
foilspfbp03	20	4.29 $\rightarrow$ 4.30	sob <i>tie</i> (0.4σ) at -32% damage

- **The $S/\sqrt{B}$ ceiling is saturated:** 4.29 $\rightarrow$ 4.30 across 74 evaluations. Three campaigns of acquisition bought 0.4σ on that axis.
- **But the front is not.** bp03 found the same significance at **32% less damage** and knocked the old record off the front entirely. A repeat campaign never raises the ceiling — it can still displace the champion *by domination*.
- Practical rule: stop when the front stops moving, **not** when the scalar record stops moving. Every jump on the sob axis itself came from changing the box (extent ceiling, then pitch), never from grinding the same one.

Conclusion & next steps

- A target optimized **at the deployed foil pitch** gives $+32\% S/\sqrt{B}$ over the deployed design, inside an envelope the existing support structure was built for.
- **No trade needed at the top any more.** The record beats the 6D champion on *both* objectives ($+4.9\% S/\sqrt{B}$, -19% damage). Against the *deployed* target it is $+32\% S/\sqrt{B}$ for $+22\%$ damage.
- **A no-cost option exists.** bp04R00_02: $+19\% S/\sqrt{B}$ at -13% damage — strictly better than the deployed target on both objectives, at the deployed foil pitch. This is the point to take to full simulation first.
- **Needed — prove it with full simulation.** Every number here is a fast score ($S/\sqrt{B}$ and damage extracted without full reconstruction). The best configs must go through the full dig $\rightarrow$ mcs $\rightarrow$ nts chain before this is an $R_{\mu e}$ claim.
- **Regime caveat.** Run-1a is cosmic-dominated, so part of the gain is window-economics. Shape gains (stopping $\times$ acceptance) survive any background model; length gains do not.
- **Design lever, transferable:** damage tracks *core* aluminium ($\rho = +0.84$), not total ($+0.57$) — keep the mass, move it off the axis. **Search closed:** foilspfbp at 93 evals; a wider-box successor (foilspfbw, 20 evals) did not improve on it at the deployed damage budget.